INCH-POUND

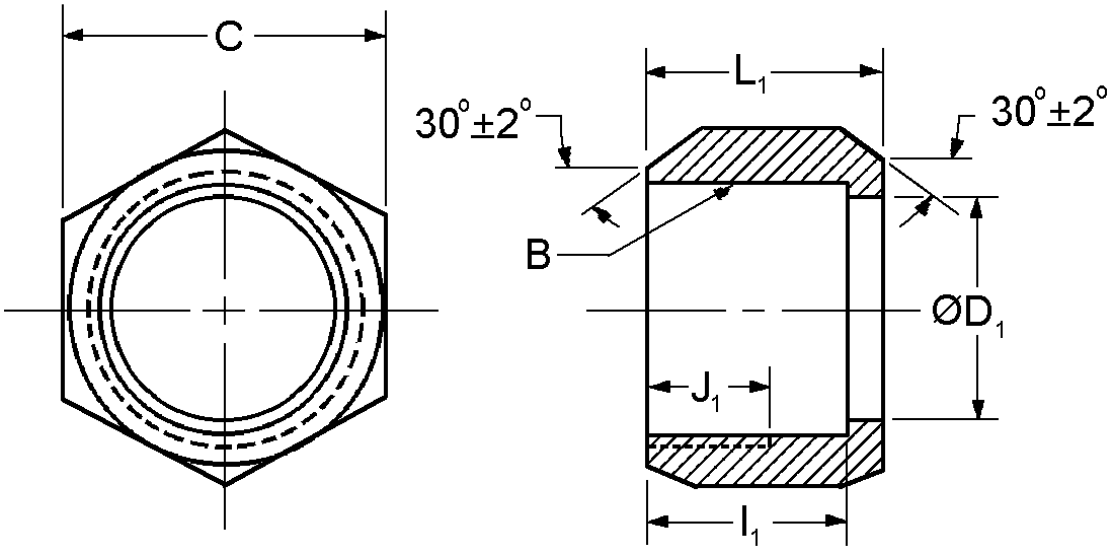
MS51531D
30 September 2021
SUPERSEDING
MS51531C
w/ AMENDMENT 1
3 February 2020

DETAIL SPECIFICATION SHEET

NUT, TUBE COUPLING, 37 DEGREE FLARED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. Nut, Tube Coupling, 37 degree flared.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

FSC 4730

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Suffix designator	Tube OD nom.	B (see note 10)	C Hex flat Nominal (See note 7)	D_1 diameter	
				Inches (mm)	Tolerance Inches (mm)
B2	.1250 (3.175)	.3125-24 UNF-2B	3/8	.179 (4.55)	+.004 (0.10) -.000
B3	.1875 (4.763)	.3750-24 UNF-2B	7/16	.241 (6.12)	
B4	.2500 (6.350)	.4375-20 UNF-2B	9/16	.304 (7.72)	
B5	.3125 (7.936)	.5000-20 UNF-2B	5/8	.373 (9.47)	
B6	.3750 (9.525)	.5625-18 UNF-2B	11/16	.439 (11.15)	
B8	.5000 (12.700)	.7500-16 UNF-2B	7/8	.568 (14.43)	+.005 (0.13) -.000
B10	.6250 (15.875)	.8750-14 UNF-2B	1	.696 (17.68)	
B12	.7500 (19.050)	1.0625-12 UN-2B	1-1/4	.832 (21.13)	
B14	.8750 (22.225)	1.1875-12 UN-2B	1-3/8	.959 (24.36)	
B16	1.0000 (25.400)	1.3125-12 UN-2B	1-1/2	1.087 (27.61)	
B20	1.2500 (31.750)	1.6250-12 UN-2B	2	1.344 (34.14)	+.006 (0.15) -.000
B24	1.5000 (38.100)	1.8750-12 UN-2B	2-1/4	1.614 (41.00)	
B32	2.0000 (50.800)	2.500-12 UN-2B	2-7/8	2.164 (54.97)	

Suffix designator	I_1 ±.010 Inches (mm)	J_1 Full thread min Inches (mm)	L_1 ±.020 (0.51) Inches (mm)
B2	.460 (11.68)	.246 (6.25)	.550 (13.97)
B3	.506 (12.85)	.277 (7.04)	.610 (15.49)
B4	.532 (13.51)	.295 (7.49)	.620 (15.75)
B5	.579 (14.71)	.326 (8.28)	.680 (17.27)
B6	.603 (15.32)	.328 (8.33)	.730 (18.54)
B8	.723 (18.36)	.370 (9.40)	.850 (21.59)
B10	.817 (20.75)	.456 (11.58)	.980 (24.89)
B12	.868 (22.05)	.464 (11.79)	1.030 (26.16)
B14	.914 (23.22)	.511 (12.98)	1.090 (27.69)
B16	.962 (24.43)	.558 (14.17)	1.130 (28.70)
B20	1.017 (25.83)	.558 (14.17)	1.230 (31.24)
B24	1.170 (29.72)	.656 (16.66)	1.420 (36.07)
B32	1.462 (37.1)	.839 (21.31)	1.750 (44.45)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges; remove all burrs, and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514 for 37° flared fittings.
5. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
6. Threads shall be in accordance with FED-STD-H28/2. Acceptability of screw threads shall be in accordance with FED-STD-H28/10, system 21.
7. Across flat widths shall fit standard wrench openings, see ASME B18.2.2.
9. Outer Diameter is abbreviated as (OD).
10. ASME B1.1 - Unified Inch Screw Threads (UN and UNR Thread Forms)

FIGURE 1. Nut, Tube Coupling, 37 degree flared- Continued.

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REQUIREMENTS:

Tube coupling nut shall be as specified on figure 1.

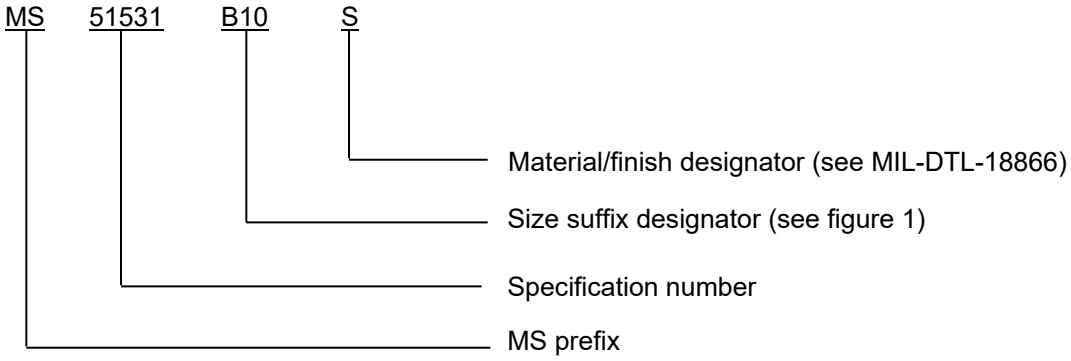
Materials shall be in accordance with MIL-DTL-18866.

Finish and finish designators shall be in accordance with MIL-DTL-18866.

Unless otherwise specified in MIL-DTL-18866, all plating's shall be capable of meeting 96-hour salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

Maximum operating pressure shall be in accordance with SAE-J514.

PIN: The PIN consists of the letters "MS", the specification number, a letter and number for nut size, and a letter for material/finish designator.



PIN example: MS51531B10S indicates a nut, tube coupling, .6250 inch (15.875 mm), corrosion resistant steel.

Marking shall be in accordance with MIL-DTL-18866.

Class I and II ozone depleting substances (ODS) shall not be used in MS51531 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

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NOTE: The following restrictions do not apply to manufacturers of the fittings and is applicable only to OEM's and users.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Army and Marine ground forces zinc plating's. PIN codes FC, FD, Z, and ZC finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army and Marine ground systems. The use of coatings depositing hexavalent chromium for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement. For new designs PIN codes FC, FD, Z, and ZC are not to be supplied to Army and Marine ground forces. MIL-PRF-32647 Type KA is intended as a hazardous free plating material and is a direct replacement for zinc plating with a hexavalent chromium conversion coating. KA has an equivalent torque/tension as the plating's replaced.

Army and Marine ground vehicles. The use of cadmium plating for Army and Marine ground vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.

Army ground vehicle applications cadmium plating. Pin code J or KB may be substituted for cadmium ("Blank" PIN code) for Army ground vehicle applications. MIL-PRF-32647 Type KB is intended as a hazardous free plating material and is a direct replacement for cadmium plating ("Blank" PIN code). KB has an equivalent torque/tension as cadmium.

NAVSEA shipboard owned equipment. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

FED-STD-H28/2
FED-STD-H28/10
MIL-PRF-32647
ASME B1.1
ASME B18.2.2
ASTM B117
SAE-J514
T9070-AL-DPC-020/077-2

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2021-032)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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